TEACHER PACKET • GEOMETRY 4TH HOUR • 45 MINUTES

Review: Midpoint, Distance & Slope

Targets: I can find the middle of a segment. I can find the length of a segment. I can find slopes and use them to compare lines.

Timeline

0–5	**Bell ringer**	ACT midpoint question on the TV (slide 2). Students work alone, then compare.
5–8	**Targets and hook**	Go over choices A, C, D (slide 2). Read the three targets (slide 3).
8–18	**I do**	Formulas (slide 4), one segment worked three ways (slide 5), add vs. subtract (slide 6).
18–28	**We do**	Problems 1–5 from the front of the student sheet (slides 7–9). Show-at-once answers.
28–40	**Partner task**	Is ABCD a square? Back of the student sheet (slides 10–11).
40–45	**Exit ticket and closure**	Exit ticket (slide 12), fist-to-five on each target, preview next class (slide 13).

Bell ringer (for edia.app)

In the standard (x, y) coordinate plane, what is the midpoint of the segment with endpoints (−3, 7) and (5, −1)?

A. (2, 6) B. (1, 3) C. (4, −4) D. (8, −8)

Answer: B

A: added but didn't divide by 2. C: half the difference. D: subtracted (that starts distance and slope, not midpoint).

Example answers (slide 5): A(−2, 1), B(4, 9)

$$M = \left(\frac{-2+4}{2}, \frac{1+9}{2}\right) = (1, 5)$$

$$d = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$$

$$m = \frac{8}{6} = \frac{4}{3}$$

Before class

- ☐ Print **1_Student_Sheets.pdf** double-sided, one per student
- ☐ Print and cut **2_Exit_Tickets.pdf**
- ☐ Enter the bell ringer in edia.app
- ☐ Open the slide deck on the TV computer
- ☐ Plan partner pairs; set out rulers
- ☐ Pull support copies of the formula box for students who need them

Watch for

- Adding when they should subtract (bell ringer C and D)
- Dropping negatives: 9 − (−3) = 12
- Problem 3: a run of 0 means the slope is undefined
- Problem 4 runs the midpoint formula backwards: model it

ANSWER KEY

We Do: Problems 1–5

1	P(−5, 2) and Q(3, 8)	$M = (-1, 5)$ $d = \sqrt{64 + 36} = 10$ $m = \frac{6}{8} = \frac{3}{4}$
2	R(0, −3) and S(5, 9)	$M = (2.5, 3)$ $d = \sqrt{25 + 144} = 13$ $m = \frac{12}{5}$
3	T(3, −2) and U(3, 6)	$M = (3, 2)$ $d = 6 - (-2) = 8$ $m = \frac{8}{0}$ → undefined (vertical)
4	M(3, −1) is the midpoint of AB; A(7, 4). Find B.	$B = (2 \cdot 3 - 7, 2 \cdot (-1) - 4) = (-1, -6)$
5	(1, 2)–(4, 8) and (0, 5)–(4, 3)	$m_1 = \frac{6}{3} = 2 \quad m_2 = \frac{-2}{4} = -\frac{1}{2}$ $2 \cdot \left(-\frac{1}{2}\right) = -1$ → **perpendicular**

Partner Task: Is ABCD a square?

A(−3, 1), B(1, 4), C(4, 0), D(0, −3)

Side	Length	Slope
AB	$\sqrt{4^2 + 3^2} = 5$	$\frac{3}{4}$
BC	$\sqrt{3^2 + 4^2} = 5$	$-\frac{4}{3}$
CD	$\sqrt{4^2 + 3^2} = 5$	$\frac{3}{4}$
DA	$\sqrt{3^2 + 4^2} = 5$	$-\frac{4}{3}$

Diagonals: both $\overline{AC}$ and $\overline{BD}$ have midpoint $\left(\frac{1}{2}, \frac{1}{2}\right)$, so they bisect each other.

Square? Yes. All four sides are 5. Opposite sides have equal slopes (parallel), and $\frac{3}{4} \cdot \left(-\frac{4}{3}\right) = -1$, so every corner is a right angle.

Done early:
$AC = BD = \sqrt{50} = 5\sqrt{2} \approx 7.07$

ANSWER KEY

Exit Ticket

#	Question	Answer
1	Midpoint of (−6, 3) and (2, −7)	$\left(\frac{-6+2}{2}, \frac{3+(-7)}{2}\right) = (-2, -2)$
2	Distance between (1, 1) and (4, 5)	$\sqrt{3^2+4^2} = \sqrt{25} = 5$
3	Slope through (−2, 5) and (2, −3)	$\frac{-3-5}{2-(-2)} = \frac{-8}{4} = -2$

Sorting the slips

Pile	Next step
3 of 3 correct	Ready for the 6.4 discovery.
Missed #1 (midpoint)	Quick midpoint check before the 6.4 discovery: the discovery starts by finding midpoints.
Missed #2 or #3	Pair them with a 3-of-3 student in the discovery so the slope and length steps don't stall.
2 or more missed	Pull for a 5-minute reteach of the three formulas at the start of next class.